COURSE INFORMATION

Clear Form | Copy Sheet | Update Index

School/Faculty:	Computing/Engineering		
Program name:	Bachelor in Software Engineering with Honors		
Course code:	SCSE2133	**Academic Session/Semester:**	2025/26/1
Course name:	Software Process and Project Management	**Pre/co requisite (course name and code, if applicable):**	SCSE1203 Software Engineering Principle
Credit hours:	3		

Course synopsis	This course provides an intermediate-level introduction to Software Project Management (SPM) with a focus on agile methodologies. It covers fundamental concepts, challenges, and management controls in SPM, followed by agile principles such as project identification, analysis, planning, documentation, and user stories. Students explore frameworks including the Dynamic Systems Development Method (DSDM), Scrum, Kanban, and Scrumban, along with project management tools and practices like MoSCoW prioritization, timeboxing, and daily stand-ups. The DSDM life cycle from pre-project to post-project phases is examined in detail, alongside agile planning, control, and testing concepts. Taught through lectures, hands-on labs, and industry-based case studies, the course emphasizes collaboration, problem-solving, and project-driven learning. By the end, students will be able to analyze and apply SPM principles, design agile project plans, evaluate and adapt agile frameworks, and implement effective prioritization and time management strategies to deliver successful software projects.
Course coordinator (if applicable)	Dr. Adila Firdaus Binti Arbain

Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail**
	Dr. Adila Firdaus Binti Arbain			adilafirdaus
	PM Dr. Rohayanti Hasan			rohayanti
	PM Dr. Hishammuddin bin Asmuni			hishamudin

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	CLO*	PLO ** (MQF Cluster Code)	***Taxonomies and ****Graduate attributes	T&L methods	*****Assessment methods
CLO1	Apply the concept correctly for software project planning and controlling in software process and project management.	PLO1 (C1)	(SC1)	Active Learning	Q, T, F
CLO2	Classify proficiently work-based projects, including effective planning and control.	PLO2 (C2)	C5 (SC2)	Active Learning, Project Based Learning, Work Based Learning	F, Pr 1, Pr 2, Pr 3
CLO3	Demonstrate rapport, interact and work effectively with others to accomplish work-based project	PLO8 (C3F)	A5 A(CC2)	Active Learning, Project Based Learning Work Based Learning	Pr 1, Pr 2, Pr 3, Pr 4
CLO4	Demonstrate entrepreneurial mindset in enhancing work-based projects	PLO10 (C4B)	A4 A (ES5)	Active Learning, Project Based Learning Work Based Learning	Pr 4

This is the basic mapping required for the CI. Any added information is allowed (extra columns for weight or other elements) provided this is made consistent for all CI at program/school/faculty level.
*Up to 5 CLO
Refer ***Taxonomies of Learning and ****UTM's Graduate Attributes for measurement of outcomes achievement
*****T – Test; Q – Quiz; Pr – Project; F – Final Exam etc.

****MQF Cluster Code**
C1 = Knowledge & Understanding, **C2** = Cognitive Skills, **C3A** = Practical Skills, **C3B** = Interpersonal Skills, **C3C**= Communication Skills, **C3D** = Digital Skills, **C3E** = Numercy Skills, **C3F** = Leadership, Autonomy & Responsibility, **C4A** = Personal Skills, **C4B** = Enterpreneurial Skills, **C5** = Ethics & Professionalism

Details on Innovative T&L practices:

No.	Type	Implementation
1	Active learning	Conducted through in-class activities
2	Project-based learning	Conducted through case study assignment. Tasks are given in sequential steps throughout the semester. Students in a group of 4 are given a case study that require a software project management planning and controlling to produce the required planning document. The report must comply to the case study specifications and be given in the form of written report.
3	Work Based Learning	Industry partner will involve in the three weeks of lecture

Transferable skills (generic skills learning in course of study which can be useful and utilised in other settings) :

Able to show leadership in team settings in diverse learning and working communities
Able to demonstrate entrepreneurial mindset in enhancing work-based projects

Student learning time (SLT) / Effective Learning Time (ELT) details:

Week/ Meeting	Course Content Outline and Subtopics	CLO* Compulsory	Learning and Teaching Activities										TOTAL SLT
			Face-to-Face (F2F)								Non F2F Independent Learning		
			Physical				Online (Synchronous)				Online (Asynchronous)	Others	
			L	T	P	O	L	T	P	O			
Week 1	C1-Introduction to Software Project Management (SPM) -The fundamental concepts of SPM -The challenges in SPM -The concepts of management control in SPM	CLO1	3								1	3	7
Week 2	C2-Software Agile Project Management Principle -The fundamental and application of agile project principle -The concepts of project identification, analysis, treatment and monitoring -Documenting planning -User stories	CLO1	3								1	3	7
Week 3	C3-Software Agile Project Management Composition -Introduction to Dynamic Systems Development Method (DSDM) Process -The composition of DSDM -DSDM and Scrum -Comparison among Scrum, Kanban, and Scrumban -Introduction to Project Management Tools	CLO1	3								1	3	7
Week 4	C4-Dynamic Systems Development Method Practices -Overview -Pre-Project Phase -Feasibility Phase -Foundations Phase -Evolutionary Development Phase -Deployment Phase -Post-Project Phase -The Life cycle in Practice -Configuring DSDM for Scalability and Formality Week 4 - Kick Off Assignment 1	CLO1	3								1	3	7
Week 5	C5-Software Agile Planning and Control -Introduction -DSDM Team -DSDM Products -Project Planning Concepts -Testing Concepts	CLO1	3								1	3	7
Week 6 Week 7	C6-Practical application of Agile PM principles -DSDM Practice - MoSCoW Prioritisation -Introduction -The MoSCoW Rules -MoSCoW Relating to a Specific Timeframe -DSDM Practice - Timeboxing -Timebox Options -The Daily Stand-up (Scrum) -Timeboxes - The Wider Context *Note: The work-based learning(WBL) is delivered as an industry case study Week6- Submission A1and Kick off Asignment 2 Week 7 Midterm 17 November 2025 8-10pm Venue TBA	CLO2	6								2	3	11
Week 8	Mid Sem Break												0
Week 9	C7-Practical application of Agile PM principles -Other DSDM Practices -Facilitated Workshops -Modelling -Iterative Development * Note:The work-based learning(WBL) is delivered as an industry case study Week 9: A2 submission and Kick off Assignment 3	CLO2	3								1	3	7
Week 10 Week 11	C8-Software Effort and Cost Estimation -Understanding Software Effort Estimation Basics -Factors Affecting Software Development Costs -Proven Techniques for Estimating Project Budgets -Function Points (FP) -COCOMO (Constructive Cost Model)	CLO2	6								2	4	12

Week 12	C9-Entrepreneur Mindset -Monetization Strategies -Market Research and Target Audience -Scalability and Growth Week 12 A3 Submission and Kick off Assignment 4	CLO2	3								1	4	8
Week 13 Week 14	C10-Security in Software Project Management -Service Level Agreement -Integration of Information Security -Educating Team Members -Use of Security Tools -Backup and Recovery Plans Week 14 A4 Submission	CLO2	6								1	4	11
Week 15	Project Presentation	CLO1 CLO2 CL04	3								1	4	8
											SUB-TOTAL SLT :		**92**

	Continous Assessment	**%**	**Face-to-Face (F2F)**		**NF2F Independent Learning for Assessment**		**TOTAL SLT**
			Physical	**Online (Synchronous)**	**Online (Asynchronous)**	**Others**	
1	Quiz 1	5	1				1
2	Project Assignment 1- User Stories	15				3	3
3	Project Assignment 2 - Moscow Classification	10				3	3
4	Project Assignment 3 - Time- Boxing	15				3	3
5	Project Assignment 4 - Monetization Planning	10				3	3
6	Mid Semester exam	15	2			4	6
					SUB-TOTAL SLT :		**19**

	Summative Assessment	**%**	**Face-to-Face (F2F)**		**NF2F Independent Learning for Assessment**		**TOTAL SLT**
			Physical	**Online (Synchronous)**	**Online (Asynchronous)**	**Others**	
1	Final Exam	30	3			6	9
2							
3							
4							
5							
					SUB-TOTAL SLT :		**9**
					SLT for Assessment:		**22**
					GRAND TOTAL SLT:		**120**

A	% SLT for F2F Physical Component	40.35
B	% SLT for Online & Independent Learning Component :	59.65
C	%SLT for Online Component:	10.83
D	% SLT for All Practical Component:	0.00
D1	% SLT for F2F Physical Practical Component:	0.00
D2	% SLT for F2F Online Practical Component:	0.00
Please tick (/) if this course is Industrial Training/ Clinical Placement/ Practicum using 50% of Effective Learning Time (ELT)		

Identify special requirement or resources to deliver the course (e.g.,software,nursery, computer lab, simulation room etc)
Project management tools **Document versioning tools**

References (include required and further readings, and should be the most current)

Stephen Grey, " Practical Risk Assessment for Project Management", Wiley (2020)
Guide to Software Project Management — Gerard O'Regan (Springer, 2025).
DSDM project Agile Project Framework- Agile Business Consortium (2014)

Other additional information (if applicable)

Academic honesty and plagiarism: *(Below is just a sample)*

Attendance is compulsory and will be taken in every lecture session. Student with less than 80% of total attendance is not allowed to sit for final exam.Students are required to behave and follow the University's dressing regulation and etiquette all the time.
Exercises and tutorial will be given in class and some may be taken for assessment. Students who do not do the exercise will lose the coursework marks for the exercise.Assignments must be submitted on the due dates. Some points will be deducted for late submissions. Assignments submitted three days after the due date will not be accepted. Make up exam will not be given, except to students who are sick and submit medical certificate which is confirmed by UTM panel doctors. Make up exam can only be given within one week of the initial date of exam.
To pass this course, students must pass both Continuous and Final assessment.

Other additional information (if applicable)

1. To pass this course, students must pass both Continuous and Final assessment.

2. All students must adhere to the following to pass a computing course:
i. Pass both continuous assessment AND final assessment
ii. Get at least minimum pass marks following UTM regulation (40%)
- The passing mark for Continuous Assessment is 30% of continuous marks.
- The passing mark for the Final Assessment is 20% of the final marks.-
- If a student does not achieve the passing mark for continuous assessment or the passing mark for the final assessment, then the student's overall final mark will be set below the UTM passing mark.
Attendance is compulsory and will be taken in every lecture session. Student with less than 80% of total attendance is not allowed to sit for final exam.

3. Students are required to behave and follow the University's dressing regulation and etiquette all the time.

4. Exercises and tutorial will be given in class and some may be taken for assessment. Students who do not do the exercise will lose the coursework marks for the exercise.

5. Assignments must be submitted on the due dates. Some points will be deducted for late submissions.
Assignments submitted three days after the due date will not be accepted.

6. Make up exam will not be given, except to students who are sick and submit medical certificate confirmed by UTM panel doctors. Make up exam can only be given within one week of the initial date of exam.

Disclaimer:

No one is allowed to use texts or excerpts from lectures or other teaching and learning activities at Universiti Teknologi Malaysia except for the purpose of his/her studies. In particular, making copies of the texts or excerpts in any form at all for the purpose of publication or distribution is strictly forbidden.While every effort has been made to ensure the accuracy of the information supplied herein, Universiti Teknologi Malaysia cannot be held responsible for any errors or omissions.

ELT = (Theory + Industrial Guidance + Assessment) x 50%

Total of credit for LI/Practical = ELT/40 Notional Hours
Note: For ODL Programme : Courses with mandatory practical requirement imposed by programme standards or any related standards can be exempted from complying to the minimum 80% ODL delivery rule in the SLT.

Prepared by:	**Certified by:**
Name: Dr. Adila Firdaus Binti Arbain	**Name:** PM Dr. Radziah Mohamad
Signature:	**Signature:**
Date: 2nd Oct 2025	**Date:**